

Rodrigo Fernandez

Senior Node.js (Express/NestJS) Backend Engineer

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Summary

Senior **Node.js Backend Engineer** with **12+ years** building SaaS, cloud infrastructure, eCommerce, and developer-platform systems using **Node.js 14.x–22.x**, **TypeScript 4.x–5.x**, **Express 4.x–5.x**, **NestJS 8.x–10.x**, **PostgreSQL 12.x–16.x**, **MongoDB 4.x–7.x**, **Redis 6.x–7.x**, and **AWS**. Specialized in secure API architecture, event-driven services, performance tuning, legacy migration, CI/CD automation, and production debugging, with strong ownership of systems that reduced latency, improved uptime, stabilized billing flows, protected customer data, and supported high-volume workflows across regulated business environments at scale.

Skills

- **Backend:** **Node.js 14.x–22.x**, **TypeScript 4.x–5.x**, JavaScript ES6+, **Express 4.x–5.x**, **NestJS 8.x–10.x**, REST API, GraphQL, WebSocket, Microservices, Serverless APIs
- **Database:** **PostgreSQL 12.x–16.x**, **MongoDB 4.x–7.x**, MySQL 8.x, Redis 6.x–7.x, Elasticsearch 7.x–8.x, Prisma 4.x–5.x, TypeORM 0.3.x, Sequelize 6.x
- **Cloud & DevOps:** **AWS**, EC2, Lambda, S3, RDS, ECS, API Gateway, CloudWatch, Docker, Kubernetes, GitHub Actions, GitLab CI, Jenkins
- **Architecture:** Event-driven architecture, message queues, CQRS, caching, rate limiting, API gateway design, authentication, authorization, payment workflows
- **Testing & Quality:** Jest 27.x–29.x, Mocha, Supertest, Pact, ESLint, SonarQube, integration testing, load testing, production debugging
- **Security:** JWT, OAuth 2.0, OpenID Connect, RBAC, OWASP API Security, encrypted secrets, audit logging, dependency scanning

Experience

Lightedge — Senior Software Engineer *(Apr 2020 – Mar 2026)*

- Built enterprise backend services with **Node.js 16.x–22.x**, **TypeScript 4.x–5.x**, and **NestJS 8.x–10.x** for cloud infrastructure platforms handling customer provisioning, billing, monitoring, and support workflows.
- Designed secure REST and event-driven APIs with **PostgreSQL 13.x–16.x**, **Redis 6.x–7.x**, and **AWS** services, improving internal service response time by **38%** through query tuning, caching, and connection pool optimization.
- Migrated legacy JavaScript services from callback-heavy **Node.js 12.x** patterns into typed **Node.js 18.x** modules, reducing production runtime errors by **42%** and making onboarding easier for new engineers.
- Resolved a recurring memory leak in a background job service by profiling heap snapshots, fixing unbounded Redis payload growth, and adding safe retry limits for failed queue messages.
- Implemented tenant-aware authorization using **OAuth 2.0**, JWT claims, and role-based access controls, strengthening account isolation for enterprise clients with complex permission models.
- Rebuilt a slow billing reconciliation workflow into smaller **NestJS** workers and asynchronous events, reducing nightly processing time by **55%** without changing the finance team's reporting flow.
- Improved API reliability by adding circuit breakers, request timeouts, structured logging, and CloudWatch alarms, helping the platform maintain stronger uptime during traffic spikes.

- Created reusable backend modules for authentication, audit trails, feature flags, and notification delivery, allowing multiple product teams to ship new services with consistent standards.
- Fixed intermittent database deadlocks in high-volume provisioning flows by redesigning transaction boundaries, adding idempotency keys, and separating read-heavy operations from write locks.
- Implemented CI/CD pipelines with GitHub Actions, Docker, automated test gates, and environment-specific secrets, reducing failed deployments by **31%** across staging and production.
- Partnered with frontend, DevOps, and security teams to deliver customer-facing infrastructure features while keeping backend services flexible enough for future cloud integrations.
- Mentored engineers on **Node.js**, **TypeScript**, API design, debugging, and production incident handling, raising code review quality and reducing repeated architecture mistakes.

NIX United — Software Engineer *(Oct 2016 – Mar 2020)*

- Developed backend platforms for SaaS and eCommerce clients using **Node.js 10.x–14.x**, **Express 4.x**, **MongoDB 4.x**, **PostgreSQL 11.x–13.x**, and message-based service integrations.
- Built REST APIs for order management, payment capture, refunds, customer profiles, and admin dashboards, improving checkout workflow stability by **34%** after refactoring fragile validation logic.
- Migrated several legacy PHP-backed endpoints into **Node.js Express** services, keeping backward-compatible API contracts while gradually moving business logic into modern service layers.
- Solved a production issue where duplicate payment webhooks created repeated invoices by adding idempotency checks, webhook signature verification, and database-level unique constraints.
- Created background workers for email, reporting, synchronization, and inventory updates using Redis queues, reducing direct API blocking time by **47%** during peak usage.
- Improved database performance by rewriting heavy aggregation queries, adding compound indexes, and separating reporting reads from transactional write paths in PostgreSQL.
- Implemented secure authentication with JWT refresh flows, password reset tokens, and scoped permissions, reducing account-related support tickets by **26%** for one client platform.
- Built integration layers for Stripe, PayPal, Shopify, and internal ERP systems, handling retry logic, failed callbacks, and data mismatch recovery with clear audit records.
- Upgraded older **Node.js 8.x–10.x** projects to **Node.js 14.x**, replacing deprecated packages, fixing dependency conflicts, and stabilizing build pipelines.
- Used Jest, Supertest, and contract testing to catch API regressions earlier, helping distributed teams release backend changes with fewer late-stage defects.
- Worked flexibly across backend, database, DevOps, and client communication responsibilities, translating unclear business requests into practical technical delivery plans.

Webiz Digital — Software Developer *(Jan 2014 – Sep 2016)*

- Built backend systems for digital commerce and marketing platforms using **Node.js 8.x–12.x**, **Express 4.x**, **MySQL 5.7–8.x**, **MongoDB 3.x–4.x**, and REST-based integrations.
- Developed APIs for product catalogs, user accounts, campaign tracking, reporting dashboards, and CMS workflows, helping clients manage content and transactions from centralized tools.
- Improved slow campaign reporting endpoints by optimizing SQL joins, adding pagination, and caching repeated dashboard queries, reducing average load time by **44%**.
- Fixed a recurring file upload failure caused by large request payloads by adding streaming uploads, validation limits, and safer error handling around cloud storage responses.
- Migrated older jQuery-driven admin workflows into API-first backend services, making frontend modernization easier without breaking existing business operations.
- Implemented role-based admin permissions, activity logs, and secure password reset flows, giving non-technical client teams safer control over sensitive operational features.

- Integrated payment, CRM, email marketing, and analytics providers through resilient **Node.js** connectors with retry logic and clearer failure notifications.
- Reduced production bug volume by **29%** by adding automated API tests, shared validation utilities, and stricter code review rules around request payload handling.
- Supported Linux server deployments, Nginx routing, PM2 process management, and basic CI scripts, keeping small-business platforms stable with limited infrastructure budgets.
- Collaborated directly with designers, project managers, and client stakeholders to turn changing requirements into backend features that stayed maintainable over time.

Aurio — Junior Software Developer *(Oct 2012 – Dec 2013)*

- Supported early-stage backend development using **JavaScript**, **Node.js 0.10.x–6.x**, **Express 3.x–4.x**, MySQL, MongoDB, and REST API patterns for web applications.
- Built basic CRUD services, authentication endpoints, admin APIs, and reporting utilities while learning production coding standards through code reviews and release support.
- Fixed form submission bugs, validation failures, and session-handling issues by tracing request logs, reproducing browser behavior, and improving server-side checks.
- Helped migrate small internal tools from mixed server-rendered logic into cleaner API routes, making the codebase easier to test and maintain.
- Improved slow database pages by adding indexes, removing unnecessary queries, and simplifying data loading flows, reducing page delays by **25%** in internal tools.
- Created reusable helpers for input validation, email notifications, date formatting, and error responses, helping the team reduce repeated backend code.
- Supported deployment troubleshooting on Linux servers by checking logs, restarting services, reviewing environment variables, and documenting repeatable fixes.
- Worked with senior engineers to understand API security, password hashing, session storage, and safe handling of customer data in real production applications.
- Gained flexible experience across backend coding, debugging, database changes, documentation, and direct support for urgent client issues.

Microsoft — Software Developer Intern *(Jun 2012 – Sep 2012)*

- Supported internal engineering tools using **C#**, **.NET Framework 4.0–4.5**, **ASP.NET MVC 3–4**, JavaScript, jQuery, and SQL Server, gaining practical experience in enterprise software quality and structured development workflows.
- Assisted senior engineers with debugging UI defects, validation issues, and data-handling problems, documenting root causes clearly so fixes could be reviewed and released with lower regression risk.
- Built small internal web modules for tracking records, reviewing status changes, and improving team visibility, applying clean HTML, CSS, JavaScript, and backend integration patterns.
- Resolved early-stage bugs related to inconsistent form validation and SQL query results by reproducing issues locally, checking logs, and working with mentors to apply safer input-handling rules.
- Learned professional engineering practices around code reviews, source control, test cases, documentation, and release discipline, which later became the foundation for stronger **React.js** and full-stack delivery.

Education

Texas State University

Bachelor's Degree in Computer Science *(Sep 2008 – May 2012)*